Unit: 6 Unit 5: Linear Equations
Topic: Parallel Lines

Name: ______________
Date: __________

Advanced (Habanero)

Q1. What is the slope of a line parallel to $y = 2x + 3$?
(A) $m = 2$
(B) $m = -2$
(C) $m = 1$
(D) $m = -1$
(E) $m = 0$

Q2. If $y = 3x - 2$ and $y = 3x + 1$, which lines are parallel?
(A) Neither
(B) Both
(C) Only the first line
(D) Only the second line
(E) They are the same line

Q3. What is the equation of a line parallel to $y = x - 4$ that passes through $(2, 3)$?
(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = x + 5$
(D) $y = -x + 5$
(E) $y = x - 5$

Q4. If two lines have slopes m_1 and m_2, what condition must be met for them to be parallel?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 =$
rac1m$_2$
(D) $m_1 = -$
rac1m$_2$
(E) $m_1 + m_2 = 0$

Q5. Which of the following lines are parallel to $y = -4x + 2$?
(A) $y = -4x - 3$
(B) $y = 4x + 1$
(C) $y = -4x + 1$
(D) $y = x - 2$
(E) $y = 2x + 5$

Q6. What can be said about the relationship between the y-intercepts of two parallel lines?
(A) They are equal
(B) They are unequal
(C) One is zero
(D) One is negative
(E) They have the same sign

Q7. If $y = mx + b$, what is the equation of a line parallel to this line that passes through $(1, 2)$?
(A) $y = m(x - 1) + 2$
(B) $y = mx + (2 - m)$
(C) $y = m(x + 1) - 2$
(D) $y = mx - (2 - m)$
(E) $y = mx + 2$

Q8. If two parallel lines have equations $y = 2x + b_1$ and $y = 2x + b_2$, what is the relationship between b_1 and b_2?
(A) $b_1 = b_2$
(B) $b_1 eq b_2$
(C) $b_1 = -b_2$
(D) $b_1 = 2b_2$
(E) $b_1 = b_2 + 1$

Open Ended Questions (FRQ)

Q9. Prove that two lines with slopes m_1 and m_2 are parallel if and only if $m_1 = m_2$. Show all work.

Q10. Find the equation of a line parallel to $y = 3x - 2$ that passes through the point $(4, 1)$. Show all work and box your answer.

Chef's Tips

- Tip 1: Make sure to check the slope of each line before determining if they are parallel.
- Tip 2: Use the point-slope form of a line, $y - y_1 = m(x - x_1)$, to find the equation of a line given the slope and a point.
- Tip 3: When proving that two lines are parallel, be sure to show that their slopes are equal.